A Benchmark for Sinhala-Script Language Identification: Sinhala, Pali, and Sanskrit

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Abstract

Most existing language identification (LangID) systems assume that text written in the Sinhala script belongs exclusively to the Sinhala language. However, the Sinhala script is extensively utilized to transcribe Pali and Sanskrit in historical, religious, and scholarly publications. This assumption leads to systematic misclassification, degrading downstream NLP tasks like OCR and machine translation. In this study, we propose the first dedicated LangID benchmark designed to differentiate Sinhala, Pali, and Sanskrit texts written entirely within the Sinhala script. In Phase 1, we evaluate baseline ML models, character-level neural networks, and multilingual transformers under full-sentence and fragment stress-testing (1, 3, and 5 words). In Phase 2, we evaluate six state-of-the-art model families zero-shot and fine-tuned across three hybrid benchmarks (FLORES+, CommonLID, WiLI-2018) spanning 11 target and global background languages. Empirical results demonstrate that character n-gram probabilistic models excel on short sub-word text fragments, while fine-tuned specialist models eliminate script ambiguity and achieve over 0.97 Macro-F1 across target and global background languages.

1 Introduction

Language Identification (LangID) serves as a critical preprocessing component in multilingual Natural Language Processing (NLP) pipelines. While modern LangID models exhibit high accuracy for globally dominant languages, they often rely heavily on orthographic features, functioning under the assumption of a bijective mapping between script and language (Joulin et al., 2017; Kargaran et al., 2023). This assumption fails catastrophically when a single script is employed to transcribe multiple distinct languages.

In Sri Lanka, the Sinhala script is the standard orthography not only for Sinhala, but also for liturgical and scholarly Pali and Sanskrit texts. Because existing tools associate the Sinhala script exclusively with the Sinhala language, they systematically fail to identify Pali and Sanskrit. This deficiency poses a significant barrier for downstream digital humanities research and NLP applications dealing with historical documents.

To address this gap, this study introduces four primary contributions across two investigation phases:

1. A comprehensive **Sinhala-Script LangID Benchmark Dataset** encompassing Sinhala, Pali, and Sanskrit written entirely within the Sinhala script.

2. **Phase 1 Baseline & Fragment Analysis**: A rigorous comparison of traditional ML (MNB, SVM), character-level neural networks (Char-CNN, Char-BiGRU), and transformers subjected to 1-, 3-, and 5-word fragment stress testing.

3. **Phase 2 State-of-the-Art Evaluation**: Construction of three hybrid benchmark datasets (FLORES+, CommonLID, WiLI-2018) combining global background languages with target test data across 11 distinct language classes.

4. **Phase 2 Adaptation & Catastrophic Forgetting Mitigation**: Benchmarking six state-of-the-art model families and engineering mitigation strategies (Two-Stage Specialist Routing, LoRA Parameter-Efficient Adaptation, C++ Causal Weight Preservation, and Head Expansion) to prevent background language degradation.

2 Dataset Construction

To train and evaluate the proposed models, a unified, sentence-level dataset was compiled. Combining multiple sources was necessary to mitigate source-specific lexical biases:

- **SiPaKosa & SiDiaC-v.2.0**: Publicly available corpora providing a foundation for Sinhala and Pali.
- **Historical & Religious Publications**: Curated supplementary texts from open digital libraries and religious commentaries.
- **Transliterated Sanskrit Data**: Due to a scarcity of native Sanskrit digitized in the Sinhala script, the Aksharamukha Asian Script Converter (Rajan, 2023) was utilized to algorithmically transliterate classical Sanskrit texts from SLP1 Romanization into the Sinhala script.

A strict group-stratified dataset split was enforced. Sentences originating from the same document, chapter, or text source were grouped together during splitting to guarantee zero data leakage across partitions. The complete target dataset comprises 67,332 sentences across Sinhala, Pali, and Sanskrit in the Sinhala script, partitioned into:

- **Training Set** (54,256 rows / 80.6%): Supervised training, character n-gram extraction, and neural parameter fine-tuning.
- **Validation Set** (6,029 rows / 9.0%): Validation loss monitoring, hyperparameter optimization, and early stopping.
- **Held-Out Test Set** (7,047 rows / 10.4%): Strictly isolated for fragment stress testing and hybrid benchmark evaluations.

2.1 Phase 2 Hybrid Benchmarking Datasets

In practical deployments, a classifier must distinguish target Sinhala-script languages from global background text (e.g., English, Tamil, German, Devanagari Sanskrit). In standard multilingual benchmarks (FLORES+, CommonLID, WiLI-2018), text written in the Sinhala script is universally tagged under a single generic Sinhala label (`sin_Sinh`, `sin`, or `si`), while Pali and Sanskrit in Sinhala script are entirely absent.

Our benchmarking taxonomy adheres strictly to the BCP-47 language tag standard (Phillips and Davis, 2009), formatting each class as `[Language]-[Script]`:

- **Target Sinhala-Script Classes**: `Sinh-Sinh`, `Pali-Sinh`, and `San-Sinh`.
- **Global Background Classes**: `San-Deva`, `Eng-Latn`, `Tam-Taml`, `Hin-Deva`, `Ben-Beng`, `Ara-Arab`, `Fre-Latn`, and `Ger-Latn`.

To construct the hybrid benchmark datasets, original generic Sinhala rows were removed from official benchmark corpora while Devanagari Sanskrit (`san_Deva`) was explicitly preserved alongside global background languages. These were replaced with our 7,047 expert-annotated target test rows:

1. **FLORES+** (Goyal et al., 2022): 229,687 rows (1,012 `san_Deva` + 7,047 target rows + FLORES+ background languages).
2. **CommonLID** (Dunn, 2023): 380,277 rows (895 `san_Deva` + 7,047 target rows + CommonLID background languages).
3. **WiLI-2018** (Thoma, 2018): 241,047 rows (1,000 `san_Deva` + 7,047 target rows + WiLI-2018 background languages).

3 Phase 1: Baselines & Fragment Stress Testing

3.1 Zero-Shot Baseline

To quantify baseline necessity, existing state-of-the-art models were evaluated zero-shot. Both Joulin et al. (2017) (fastText) and Kargaran et al. (2023) (GlotLID) failed to identify Pali and Sanskrit, defaulting entirely to Sinhala or "other". This resulted in an aggregate Macro-F1 score of approximately 0.18, confirming that off-the-shelf solutions are inadequate.

3.2 Full Sentence Evaluation

Models trained on the curated dataset achieved near-perfect classification on full sentences (Table 1). Because full sentences provide extensive contextual and syntactic clues, both traditional machine learning models (Joachims, 1998; McCallum and Nigam, 1998) and massive transformers (Conneau et al., 2020; Devlin et al., 2019) achieved up to 1.000 Macro-F1 scores.

3.3 Fragment Stress Testing

To evaluate morphological understanding, a stress test was designed restricting input to fragments of 5, 3, and 1 randomly selected words per sentence. This constraint eliminates contextual syntax, forcing models to rely strictly on root words and affixes.

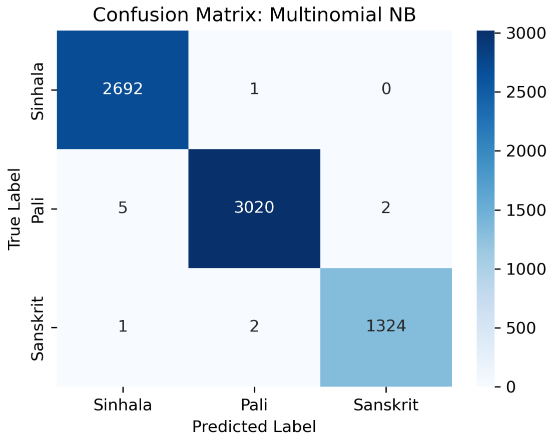

Figure 1: Confusion Matrix for Multinomial Naive Bayes.

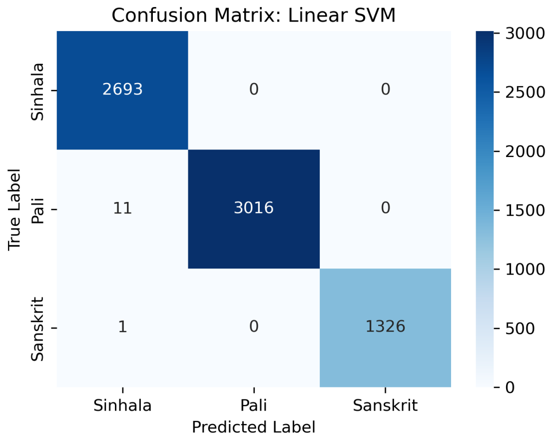

Figure 2: Confusion Matrix for Linear SVM.

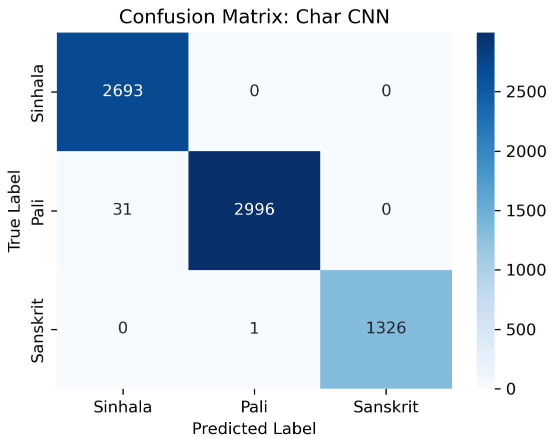

Figure 3: Confusion Matrix for Character-level CNN.

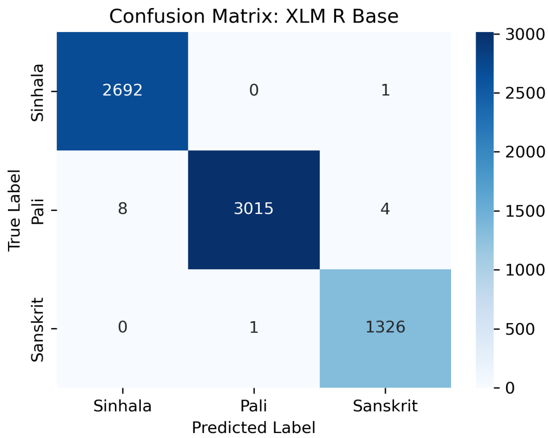

Figure 4: Confusion Matrix for XLM-R Base Transformer.

As shown in Table 2, large transformers experienced significant degradation as fragment length decreased. In contrast, character n-gram models demonstrated exceptional robustness across all fragment sizes.

4 Phase 1: Architectural Analysis

Morphological affixes (prefixes, suffixes, and orthographic character combinations) form the core distinguishing boundary between Sinhala, Pali, and Sanskrit written in the same script. Traditional models utilizing character n-grams (2 to 4 characters) with TF-IDF vectorization proved exceptionally adept at capturing these markers. Multinomial Naive Bayes (McCallum and Nigam, 1998) (Figure 1) treated n-gram frequencies probabilistically without relying on syntax. For example, the presence of the distinct Pali suffix *-ssa* allows MNB to detect a high-probability marker immediately, rendering sentence context unnecessary. Support Vector Machines (Joachims, 1998) (Figure 2) demonstrated similar robustness.

Character-level neural networks, including Char-CNN (Zhang et al., 2015) (Figure 3) and Char-BiGRU (Cho et al., 2014), autonomously learned morphological boundaries, reaching 1.00 Macro-F1 on full sentences. However, on fragment tests, their accuracy dropped slightly due to sensitivity to altered sequence padding distributions.

Large multilingual transformers (Conneau et al., 2020; Devlin et al., 2019) (Figure 4) rely on subword tokenization designed for major global languages. When input is restricted to 1-to-3 words, the transformer loses contextual syntax. Stripped of sentence structure, subword embeddings fail to discriminate between inflected morphological roots,

Model	Architecture	Acc.	M-F1
Multinomial NB	ML (Probabilistic)	0.998	0.998
Linear SVM	ML (Margin-based)	0.998	0.998
fastText (re-trained)	fastText	0.998	0.998
XLM-R LangID	Transformer	0.998	0.998
XLM-R base	Transformer	1.000	1.000
Char-CNN	Deep Learning	1.000	1.000
Char-BiGRU	Deep Learning	1.000	1.000
XGBoost	Gradient Boosting	0.995	0.995
Char n-gram + LogReg	ML (Baseline)	0.990	0.990
Word-BiGRU	Deep Learning	0.980	0.980
GlotLID + LogReg Head	Transfer Learning	0.974	0.970
Word2Vec + LogReg	Word Embeddings	0.940	0.950
mBERT (fine-tuned)	Transformer	0.940	0.940

Table 1: Phase 1 full sentence evaluation across all baseline models.

causing XLM-R's Macro-F1 to drop to 0.662 on 1-word fragments—below baseline Logistic Regression (0.680).

5 Phase 2: SOTA Benchmarking & Adaptation

Phase 2 expanded the framework across six state-of-the-art model families: XLM-R Base (Conneau et al., 2020), ConLID, fastText LID-176 (Joulin et al., 2017), GlotLID v3 (Kargaran et al., 2023), NLLB LID-218 (NLLB Team et al., 2022), and OpenLID-v2 (Burchell et al., 2023).

5.1 Mitigation of Catastrophic Forgetting

Direct fine-tuning causes severe catastrophic forgetting of pre-trained global background languages. To preserve background coverage while adding Sinhala-script Pali and Sanskrit capabilities, five strategies were engineered:

1. **Two-Stage Specialist Routing** (OpenLID-v2 & fastText LID-176): A Stage 1 router intercepts input text. Non-Sinhala scripts are output directly, while Sinhala script (`si/sin_Sinh`) is routed to a fine-tuned Stage 2 3-way classifier, guaranteeing zero degradation on background languages.

2. **LoRA Adaptation with Mixed-Language Rebalancing** (XLM-R Base): Low-Rank Adaptation (Hu et al., 2022) matrices were injected into attention layers, while global background samples were remixed into fine-tuning batches to retain pre-trained representations.

3. **Weight-Preserving Head Extension & Continuation Training** (NLLB LID-218): A custom C++ patch expanded the fastText output matrix from 218 to 220 labels while copying pre-existing weights verbatim, followed by continuation training ($lr = 0.001$, 3 epochs).

4. **In-Place Head Extension with Xavier Initialization** (ConLID): The final PyTorch linear layer was expanded with Xavier uniform initialization for new rows and trained using AdamW (lr = 1e-4) with validation early stopping.

5. **Pretrained Vector Transfer** (GlotLID v3): Pretrained subword vectors were exported to initialize training with a conservative schedule ($lr = 0.005$, 5 epochs) to restrict weight drift.

Model	Accuracy	Macro-F1
5-Word Fragments		
fastText (re-trained)	0.993	0.993
Char n-gram + LogReg	0.990	0.990
XLM-R LangID	0.983	0.979
Word2Vec + LogReg	0.920	0.910
GlotLID + LogReg Head	0.860	0.842
3-Word Fragments		
Multinomial NB	0.987	0.985
fastText (re-trained)	0.972	0.967
Char n-gram + LogReg	0.960	0.950
Linear SVM	0.941	0.934
XLM-R LangID	0.927	0.908
Word2Vec + LogReg	0.870	0.850
GlotLID + LogReg Head	0.784	0.746
1-Word Fragments		
fastText (re-trained)	0.821	0.791
Char n-gram + LogReg	0.730	0.680
XLM-R LangID	0.749	0.662
Word2Vec + LogReg	0.650	0.630
GlotLID + LogReg Head	0.638	0.549

Table 2: Phase 1 fragment stress testing performance across 5-word, 3-word, and 1-word inputs.

5.2 Benchmark Performance Overview

Table 3 summarizes overall Macro-F1 performance across all three hybrid benchmarks. Comprehensive 11-language breakdowns for FLORES+, CommonLID, and WiLI-2018 are presented in Appendix A.

6 Discussion: Devanagari Script Dialects

While target Sinhala-script languages achieved 0.96–0.98 F1 across Phase 2 models, OpenLID-v2 exhibited a significantly lower overall Macro-F1 (0.77–0.79). In OpenLID-v2, stock Stage 1

Model	FLORES+	CommonLID	WiLI-18
XLM-R Base	0.9267	**0.9737**	0.8880
ConLID	0.9329	0.9261	0.9688
fastText LID-176	0.9276	0.9645	0.9745
GlotLID v3	0.9465	0.9527	0.9570
NLLB LID-218	**0.9530**	0.9501	**0.9675**
OpenLID-v2	0.7759	0.7913	0.7782

Table 3: Overall Macro-F1 across the three Phase 2 hybrid benchmarks for adapted fine-tuned models.

splits Devanagari text across fine-grained regional dialects (Bhojpuri `bho_Deva` 68.1%, Maithili `mai_Deva` 15.1%, Kashmiri `kas_Deva` 43.3%), dropping exact single-label match F1 for `san_Deva` to 0.035. In contrast, fastText LID-176 utilizes a unified `sa` label, maintaining 0.9594 to 0.9904 F1 on `san_Deva`.

7 Conclusion

This paper presents the first dedicated LangID benchmark for Sinhala, Pali, and Sanskrit written in the Sinhala script. In Phase 1, character n-gram probabilistic classifiers proved superior to transformers on short, ambiguous fragments. In Phase 2, targeted adaptation strategies resolved script ambiguity while preserving classification performance across 11 global background languages.

Limitations

The Sanskrit dataset partially relies on algorithmic transliteration via Aksharamukha rather than exclusively native historical manuscripts. Additionally, performance on heavily damaged historical palm-leaf manuscripts containing non-standard orthography and archaic character variants requires further investigation.

References

Laurie Burchell, Alexandra Birch, Nikolay Bogoychev, and Kenneth Heafield. 2023. An open-source language identification tool for low-resource languages (OpenLID). In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 160–169. Association for Computational Linguistics.

Kyunghyun Cho, Bart van Merriënboer, Caglar Gulcehre, Dzmitry Bahdanau, Fethi Bougares, Holger Schwenk, and Yoshua Bengio. 2014. Learning phrase representations using RNN encoder–decoder for statistical machine translation. In *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 1724–1734. Association for Computational Linguistics.

Alexis Conneau, Kartikay Khandelwal, Naman Goyal, Vishrav Chaudhary, Guillaume Wenzek, Francisco Guzmán, Edouard Grave, Myle Ott, Luke Zettlemoyer, and Veselin Stoyanov. 2020. Unsupervised cross-lingual representation learning at scale. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 8440–8451. Association for Computational Linguistics.

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 4171–4186. Association for Computational Linguistics.

Jonathan Dunn. 2023. CommonLID: Language identification for massively multilingual web corpora. In *Proceedings of the EACL Benchmark Workshop*.

Naman Goyal, Cynthia Gao, Vishrav Chaudhary, Peng-Jen Chen, Guillaume Wenzek, Da Ju, Sanjana Krishnan, Marc Riemersma, Francisco Guzmán, and Angela Fan. 2022. The FLORES-101 evaluation benchmark for low-resource language translation. *Transactions of the Association for Computational Linguistics*, 10:522–538.

Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*.

Thorsten Joachims. 1998. Text categorization with support vector machines: Learning with many relevant features. In *European Conference on Machine Learning*, pages 137–142. Springer.

Armand Joulin, Edouard Grave, Piotr Bojanowski, and Tomas Mikolov. 2017. Bag of tricks for efficient text classification. In *Proceedings of the 15th Conference of the European Chapter of the Association for Computational Linguistics: Volume 2, Short Papers*, pages 427–431. Association for Computational Linguistics.

Amir Hossein Kargaran, Ayyoob Imani, François Yvon, and Hinrich Schütze. 2023. GlotLID: Language identification for low-resource languages. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 8370–8397. Association for Computational Linguistics.

Andrew McCallum and Kamal Nigam. 1998. A comparison of event models for naive bayes text classification. In *AAAI-98 Workshop on Learning for Text Categorization*, volume 752, pages 41–48.

NLLB Team, Marta R. Costa-jussà, James Cross, Onur Çelebi, Maha Elbayad, Kenneth Heafield, et al. 2022. No language left behind: Scaling human-centric machine translation. *arXiv preprint arXiv:2207.04672*.

Addison Phillips and Mark Davis. 2009. Tags for identifying languages. Ietf best current practice bcp 47, rfc 5646.

Vinodh Rajan. 2023. Aksharamukha: Asian script converter. https://aksharamukha.appspot.com/.

Martin Thoma. 2018. WiLI-2018—Wikipedia language identification benchmark dataset. *arXiv preprint arXiv:1801.07779*.

Xiang Zhang, Junbo Zhao, and Yann LeCun. 2015. Character-level convolutional networks for text classification. In *Advances in Neural Information Processing Systems*, volume 28.

A Detailed Hybrid Benchmark Results

Tables 4, 5, and 6 provide the complete fine-grained performance breakdown across all 11 target and global background languages.

Model	Sinh-Sinh	Pali-Sinh	San-Sinh	San-Deva	Eng-Latn	Tam-Taml	Hin-Deva	Ben-Beng	Ara-Arab	Fre-Latn	Ger-Latn	Macro F1
Zero-Shot Baseline Results												
XLM-R Base	0.9985	–	–	0.1835	0.6671	0.9990	1.0000	–	–	0.9985	0.7916	–
ConLID	0.9960	–	–	0.9995	0.9970	0.3733	0.9995	1.0000	0.9970	–	–	–
fastText LID-176	0.7912	–	–	0.9594	0.7213	1.0000	0.9629	0.6667	1.0000	0.9873	0.9897	–
GlotLID v3	0.7914	–	–	0.9950	1.0000	1.0000	0.9926	0.9995	0.6054	1.0000	1.0000	0.9895
NLLB LID-218	0.7912	–	–	0.9840	1.0000	0.9912	0.6664	1.0000	0.9995	1.0000	–	–
OpenLID-v2	0.7912	–	–	0.0350	0.9995	1.0000	0.0361	0.5579	0.9970	0.9975	1.0000	–
Fine-Tuned Model Results												
XLM-R Base	0.9967	0.9844	0.9974	–	0.7734	0.9912	–	0.9970	0.6667	0.8604	1.0000	0.9267
ConLID	0.9634	0.9705	0.9414	0.9985	0.9960	0.9995	0.9970	0.9995	0.3991	1.0000	0.9970	0.9329
fastText LID-176	0.9611	0.9751	0.9805	0.9594	0.7213	1.0000	0.9629	1.0000	0.6667	0.9873	0.9892	0.9276
GlotLID v3	0.9688	0.9566	0.9019	0.9955	1.0000	1.0000	0.9926	0.9995	0.5962	1.0000	1.0000	0.9465
NLLB LID-218	0.9526	0.9760	0.9053	0.9915	1.0000	1.0000	0.9921	1.0000	0.6667	0.9995	0.9995	0.9530
OpenLID-v2	0.9670	0.9743	0.9704	0.0350	0.9995	1.0000	0.0361	0.9970	0.5579	0.9975	1.0000	0.7759

Table 4: FLORES+ Hybrid Benchmark Performance: Zero-Shot Baselines vs. Fine-Tuned Models across all 11 languages.

Model	Sinh-Sinh	Pali-Sinh	San-Sinh	San-Deva	Eng-Latn	Tam-Taml	Hin-Deva	Ben-Beng	Ara-Arab	Fre-Latn	Ger-Latn	Macro F1
Zero-Shot Baseline Results												
XLM-R Base	–	–	–	0.9473	0.4403	–	0.9955	0.9483	0.9677	–	–	0.9591
ConLID	0.7916	–	–	0.8500	0.9818	0.9443	0.9615	0.7664	0.8974	0.8953	–	0.8886
fastText LID-176	0.7910	–	–	0.9724	0.9702	0.9643	0.9936	0.9947	0.9447	0.9673	–	0.9634
GlotLID v3	0.7914	–	–	0.9255	0.9713	0.9878	0.9792	0.9631	0.9255	0.9215	–	–
NLLB LID-218	0.7912	–	–	0.9282	0.9240	0.9640	0.9747	0.9863	0.9923	0.9289	0.9410	–
OpenLID-v2	0.7912	–	–	0.1169	0.8973	0.1749	0.9877	0.9722	0.9473	0.9052	0.9239	–
Fine-Tuned Model Results												
XLM-R Base	0.9966	0.9901	0.9966	–	0.9786	0.9310	0.9925	0.9865	0.9971	0.9521	0.9665	0.9737
ConLID	0.9634	0.9705	0.9414	0.9607	0.8488	0.9818	0.9415	0.9592	0.7966	0.8975	0.8888	0.9261
fastText LID-176	0.9609	0.9751	0.9805	0.8886	0.9724	0.9643	0.9702	0.9936	0.9947	0.9447	0.9672	0.9645
GlotLID v3	0.9566	0.9689	0.9019	0.9612	0.9270	0.9818	0.9646	0.9792	0.9603	0.9256	0.9211	0.9527
NLLB LID-218	0.9625	0.9500	0.9053	0.9310	0.9072	0.9875	0.9588	0.9874	0.9919	0.9192	0.9407	0.9501
OpenLID-v2	0.9743	0.9670	0.9704	0.1169	0.8973	0.9877	0.1749	0.9722	0.9473	0.9052	0.9239	0.7913

Table 5: CommonLID Hybrid Benchmark Performance: Zero-Shot Baselines vs. Fine-Tuned Models across all 11 languages.

Model	Sinh-Sinh	Pali-Sinh	San-Sinh	San-Deva	Eng-Latn	Tam-Taml	Hin-Deva	Ben-Beng	Ara-Arab	Fre-Latn	Ger-Latn	Macro F1
Zero-Shot Baseline Results												
XLM-R Base	0.9527	–	–	0.1811	0.9900	0.9899	–	–	–	–	–	–
ConLID	0.7916	–	–	0.9924	0.9439	0.9945	0.9899	0.9435	0.9819	0.9707	–	–
fastText LID-176	0.7912	–	–	0.9904	0.9385	0.9950	0.9889	0.9774	0.9529	0.9854	–	–
GlotLID v3	0.7914	–	–	0.9914	0.9377	0.9950	0.9173	0.9880	0.9451	0.9754	–	–
NLLB LID-218	0.7912	–	–	0.9909	0.9301	0.9940	0.9889	0.9900	0.9429	0.9904	–	–
OpenLID-v2	0.7912	–	–	0.0276	0.9427	0.9940	0.0134	0.9424	0.9687	0.9817	–	–
Fine-Tuned Model Results												
XLM-R Base	0.9970	0.9967	0.9957	0.9569	0.9950	0.9909	0.9701	0.9886	0.9894	0.8880	–	–
ConLID	0.9634	0.9705	0.9414	0.9924	0.9433	0.9945	0.9899	0.9435	0.9809	0.9680	0.9688	–
fastText LID-176	0.9611	0.9751	0.9805	0.9904	0.9385	0.9950	0.9889	0.9529	0.9774	0.9850	0.9745	–
GlotLID v3	0.9688	0.9566	0.9019	0.9914	0.9367	0.9950	0.9126	0.9451	0.9875	0.9744	0.9570	–
NLLB LID-218	0.9760	0.9526	0.9053	0.9909	0.9455	0.9940	0.9879	0.9429	0.9900	0.9904	0.9675	–
OpenLID-v2	0.9670	0.9743	0.9704	0.0276	0.9427	0.9940	0.0134	0.9424	0.9687	0.9817	0.7782	–

Table 6: WiLI-2018 Hybrid Benchmark Performance: Zero-Shot Baselines vs. Fine-Tuned Models across all 11 languages.